

• If the edge weights are distinct, then we have a unique MST.

Assume there are two MSTs : T_1 and T_2 . (for the sake of contradiction)

$e_1 \in T_1 / T_2$
↳ min. wt. edge in T_1 / T_2

$e_2 \in T_2 / T_1$
↳ min. wt. edge in T_2 / T_1

$wt(e_1) < wt(e_2)$

$T_2 \cup \{e_1\}$ → this has a cycle

→ $e_3 \in T_2, e_3 \notin T_1 / T_2$

Remove an edge e_3 from the cycle.
 $T_3 = T_2 \cup \{e_1\} - \{e_3\}$
 $wt(T_3) = wt(T_2) + wt(e_1) - wt(e_3)$

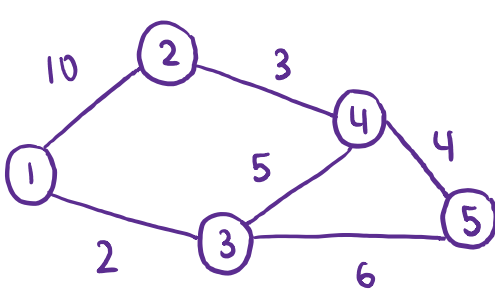
Swapping / Exchange Arguments

If $e_3 = e_2$, then $wt(T_3) < wt(T_2) \Rightarrow$ contradicts minimality of T_2
same edge ($e_3 = e_2$ also understood)

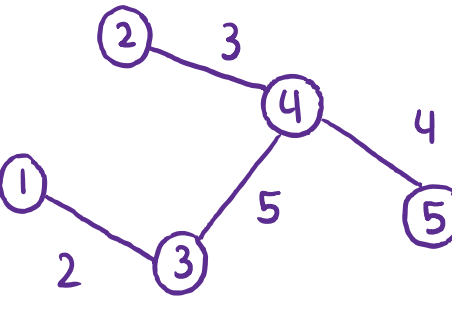
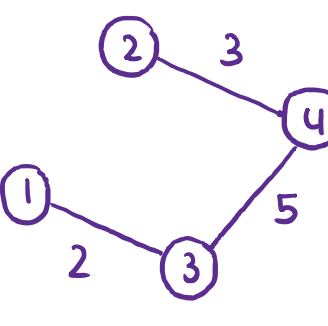
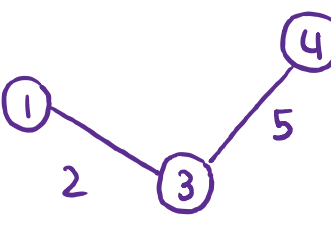
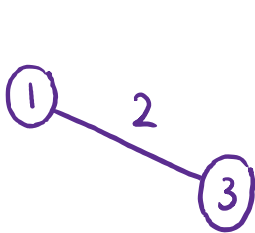
$e_3 \in T_2 / T_1$ (by enforcing that we pick an edge that does not belong to T_1 from the cycle).

• REFERENCE: Jeff Erickson's MST chapter

• PRIM'S ALGORITHM :



• Over all $N(S)$:
↳ pick v s.t. v attains min. edge wt. with a vertex in S .

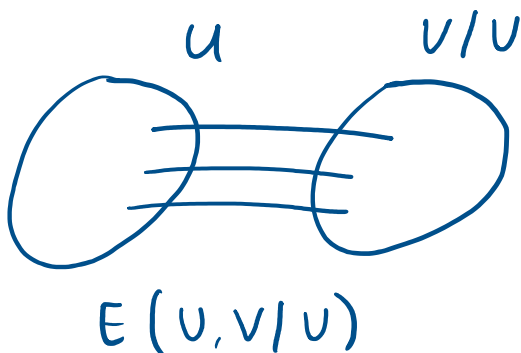


$T \leftarrow \{\}$
 $S \leftarrow \{s\}$
While $S \neq V$:
Pick v and edge (u,v) s.t.
• $v \in N(S)$ and $u \in S$
• $wt(u,v) \leq wt(u,v') \quad \forall u' \in S \text{ and } v' \in N(S) \cap V \setminus S$
 $T \leftarrow T \cup \{(u,v)\}$
 $S \leftarrow S \cup \{v\}$

n Extract mins + $O(n)$ overhead + $O(m)$

• **Cycle Property**: Let C be any cycle in the given graph.
Let edge f be the max. edge weight in C .
Then f cannot be in the MST.

• **Cut Property**: Let U be a subset of V .
Let e be the min. wt. edge in $E(U, V/U)$.
Then, an MST contains e .
sufficient to prove correctness of Prim's



PROOF:
Suppose T^* be the MST st. it contains f .
 $T^* - \{f\}$ is a disconnected graph.
(It is a forest with 2 trees)

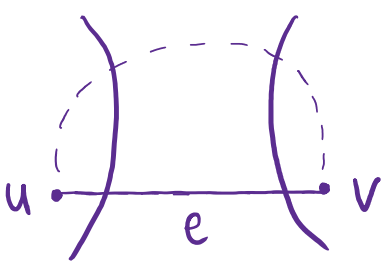
Let $e_1, e_2, \dots, e_k$ be edges in the cycle apart from f .
 $(u_1, v_1), \dots, (u_i, v_i), \dots, (u_k, v_k)$
 $\forall i$, there was a path from u_i to v_i .

There must be a pair (u_i^*, v_i^*) that are disconnected.
Adding that edge e_i^* would give us another spanning tree.

$T \leftarrow (T^* - \{f\}) \cup \{e_i^*\}$
 $wt(T) = wt(T^*) - wt(f) + wt(e_i^*) < wt(T^*)$

This contradicts the minimality of T^* .

PROOF:
Let T^* be the MST that does not contain $e = (u,v)$.
 $u \in U \quad v \in V/U$



Since T^* is a spanning tree, u and v must be connected through a path which also must have an edge that crosses the cut.

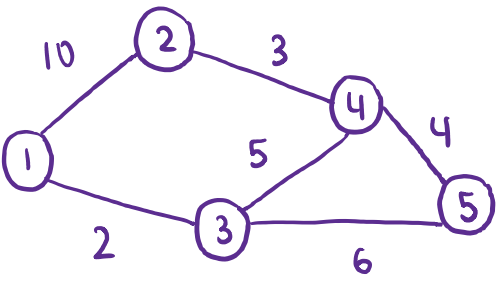
Let f be that edge.

$T \leftarrow T^* - \{f\} \cup \{e\}$
 $wt(T) < wt(T^*)$

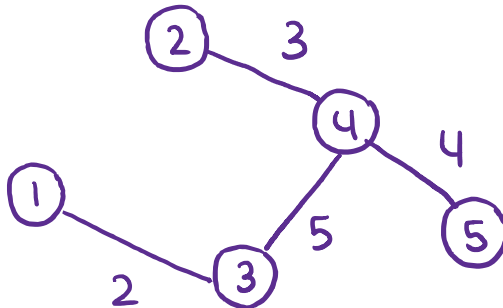
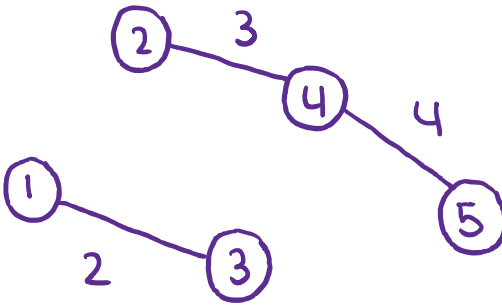
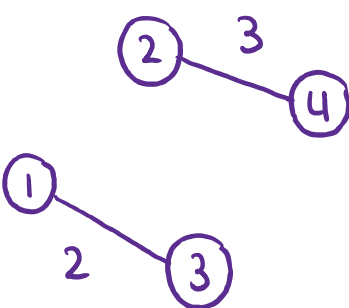
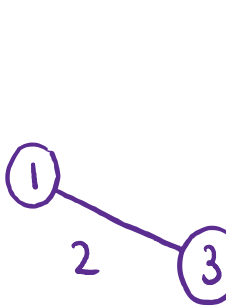
This contradicts the minimality of T^* .

• **Correctness of each step of Prim's is guaranteed by cut property.**

• KRUSKAL'S ALGORITHM:
• Sort the edges in increasing order of their weights.
• Add edges to a tree as long as they do not create cycles.



$(1,3), (2,4), (4,5), (3,4), (3,5), (1,2)$



* **Cycle property guarantees the correctness of Kruskal's algorithm.**
(by considering edges in the sorted order)

PRIM'S : cut property for correctness
KRUSKAL'S : cycle property for correctness